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# Dispatchable Virtual Oscillator Control (dVOC) for a 2L-VSI

This model builds a three-phase *grid-forming* inverter whose outer control loop is not the usual droop controller but a nonlinear *oscillator* — the Andronov–Hopf oscillator, the heart of *dispatchable Virtual Oscillator Control* (dVOC). Everything is built from plain PLECS blocks (no C-code). Background comes from (Rosso et al., 2021) and (Seo et al., 2019)

## 1 The big picture: why an oscillator?

Most inverters are *grid-following*: a PLL reads the grid angle and the inverter provides current behind that angle. That only works if the voltage is being held up (a stiff grid is required), if that's not the case, then we need inverters that can *be* the grid instead of just following it. Those are *grid-forming* (GFM) inverters: they behave like a controllable AC voltage source and let the power sort itself out from the physics of the line.

Droop control is the classic way to do this. *Virtual oscillator control* (VOC) is a newer idea: instead of computing an angle, you let a nonlinear oscillator *be* the AC waveform — its stable, self-sustaining oscillation is the output voltage. The neat result is that a bunch of these oscillators automatically lock to the same frequency and voltage, and in steady state the whole thing behaves *exactly* like droop — but unlike droop it stays stable during big transients and black starts. The *dispatchable* version (dVOC) just adds power set-points  $P^*, Q^*$  so you can command it where to go. That is what we built.

## 2 The oscillator core

### 2.1 The two equations

Picture a virtual  $LC$  tank with two states: a capacitor voltage  $v_c$  and an inductor current  $i_L$ . They evolve as

$$\begin{cases} L \frac{di_L}{dt} = v_c + v_m - \varepsilon u_2, \\ C \frac{dv_c}{dt} = -i_L + i_m - u_1. \end{cases} \quad (1)$$

Ignore  $v_m, i_m, u_1, u_2$  for a second and this is just an ideal  $LC$  circuit that rings forever at

$$\omega_g = \frac{1}{\sqrt{LC}}. \quad (2)$$

The extra terms do two jobs:  $v_m, i_m$  hold the *amplitude* in place, and  $u_1, u_2$  steer the *phase and power*. That's the whole controller.

## 2.2 Holding the amplitude: the Hopf trick

Collect the states into a vector and look at its length:

$$x = \begin{bmatrix} v_c \\ \varepsilon i_L \end{bmatrix}, \quad \|x\|^2 = v_c^2 + \varepsilon^2 i_L^2. \quad (3)$$

The amplitude terms are

$$v_m = \frac{\xi}{\omega_{\text{ref}}} \left( 2 \left( \frac{V_{\text{ref}}}{k_v} \right)^2 - \|x\|^2 \right) \varepsilon i_L, \quad (4)$$

$$i_m = \frac{\xi}{\varepsilon \omega_{\text{ref}}} \left( 2 \left( \frac{V_{\text{ref}}}{k_v} \right)^2 - \|x\|^2 \right) v_c. \quad (5)$$

For an easier understanding, assume  $k_v = 1$  (this term is not relevant for now), now, focus on the bracket  $(2V_{\text{ref}}^2 - \|x\|^2)$ , it's the clever bit. It acts like a damping term that changes sign:

- Oscillation *too small* ( $\|x\|^2 < 2V_{\text{ref}}^2$ ): bracket positive  $\Rightarrow$  it *adds* energy, amplitude grows.
- Oscillation *too big* ( $\|x\|^2 > 2V_{\text{ref}}^2$ ): bracket negative  $\Rightarrow$  it *removes* energy, amplitude shrinks.
- Exactly right ( $\|x\|^2 = 2V_{\text{ref}}^2$ ): nothing happens — a stable *limit cycle*.

So from almost any starting point the oscillation spirals out (or in) to a circle of radius  $\sqrt{2} V_{\text{ref}}$  and stays there. (This self-correcting behaviour is a supercritical *Andronov–Hopf bifurcation*, hence the name.) The knob  $\xi$  (xi) sets *how fast* it settles; roughly,

$$\tau_{\text{amp}} \approx \frac{1}{4 \xi V_{\text{ref}}^2}. \quad (6)$$

With  $\xi = 100$ ,  $V_{\text{ref}} = 1$  that's about 2.5 ms, and a black start from the tiny seed  $v_c(0) = 10^{-3}$  reaches full voltage in roughly 40 ms.

## 2.3 Why this gives you a voltage

On the limit cycle  $\|x\|^2 = 2V_{\text{ref}}^2$ , so  $v_c$  is a sine wave with peak  $\sqrt{2} V_{\text{ref}}$  — i.e. RMS value  $V_{\text{ref}}$ . So  $V_{\text{ref}}$  is just your (per-unit) RMS phase-voltage target. The pair  $(v_c, \varepsilon i_L)$  is a vector of constant length spinning at  $\omega_g$ ; its two components are the  $\alpha, \beta$  parts of the output voltage,

$$v_\alpha = k_v v_c, \quad v_\beta = \varepsilon k_v i_L, \quad (7)$$

and an inverse Clarke transform turns those into the three phase references for the PWM. No PLL, no angle integrator; the oscillator *is* the angle generator.

## 3 Telling it how much power to make: the $\Gamma$ matrix

The oscillator sets the voltage. A small algebra block then works out what current that voltage *should* be pushing to hit your power commands. Instantaneous power in the  $\alpha\beta$  frame is

$$P = \frac{3}{2} (v_\alpha i_\alpha + v_\beta i_\beta), \quad Q = \frac{3}{2} (v_\beta i_\alpha - v_\alpha i_\beta). \quad (8)$$

Solve those for the current that would deliver  $P^*, Q^*$  and you get the dispatch matrix  $\Gamma$ :

$$\boxed{\begin{bmatrix} i_\alpha^* \\ i_\beta^* \end{bmatrix} = \frac{2}{3 \|v\|^2} \begin{bmatrix} v_\alpha & v_\beta \\ v_\beta & -v_\alpha \end{bmatrix} \begin{bmatrix} P^* \\ Q^* \end{bmatrix}, \quad \|v\|^2 = v_\alpha^2 + v_\beta^2.} \quad (9)$$

## 4 Keeping it in sync: the current-error loop

Left alone the oscillator would drift, so we feed back how far the real current is from the  $\Gamma$  target. That’s what  $u_1, u_2$  are for. Measure the grid-side current (at the PCC, after the LCL filter), take the error against the reference, scale by  $k_i$ , and rotate it by an angle  $\phi$ :

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} k_i (i_\alpha - i_\alpha^*) \\ \varepsilon k_i (i_\beta - i_\beta^*) \end{bmatrix}. \quad (10)$$

This drops straight back into (1): a phase error nudges the oscillator’s angle (that’s the  $P$ /frequency channel), an amplitude error nudges its size (the  $Q$ /voltage channel).

**What  $\phi$  does.** It matches the line’s impedance angle. For a mostly *inductive* line,  $\phi = \pi/2$  cleanly separates “ $P$  moves the angle” from “ $Q$  moves the voltage” (the familiar droop split). We ship  $\phi = \pi/2$  in our application.

**What  $k_i$  does.** It’s the sync gain: bigger  $k_i$  chases the power reference faster but reacts harder to grid disturbances.

## 5 Per-unit inside, real volts outside

The oscillator runs in per-unit and only gets scaled to real volts and amps at its terminals:

- **Tank.**  $k_v = 1$ ,  $C_{\text{osc}} = 1$ ,  $L_{\text{osc}} = 1/\omega_g^2$  so the natural frequency lands on the grid frequency. The scaling factor is the tank’s characteristic impedance  $\varepsilon = \sqrt{L_{\text{osc}}/C_{\text{osc}}} = 1/\omega_g$ , which is what keeps  $x = [v_c, \varepsilon i_L]^\top$  a nice circle.
- **Voltage out.** Since the peak is  $\sqrt{2} V_{\text{ref}}$ , the output gain  $K_{v,\text{out}} = V_{\text{base, FN, peak}}/\sqrt{2} = V_{\text{base, RMS}}/\sqrt{3}$  maps  $V_{\text{ref}} = 1$  to the rated RMS phase voltage ( $\approx 63.5$  kV on a 110 kV line-to-line base).
- **Current in.** Measured currents are divided by  $I_{\text{norm}} = I_{\text{base, peak}}$ , so  $k_i$  stays a per-unit gain.

## 6 Three add-ons you can toggle

These next concepts sit around the basic dVOC. Each is there to teach one idea, and each can be switched off so you can see the “before” and “after”.

### 6.1 Pre-synchronisation (soft grid connection)

Close the breaker while the oscillator is at a random phase and you get an inrush spike. So before connecting, we push a virtual “line-up” current into the error loop,

$$i_{\text{sync}} = K_{\text{sync}} (v_{\text{inv}} - v_{\text{grid}}), \quad K_{\text{sync}} = k_{\text{sync, gain}} \frac{I_{\text{norm}}}{V_{\text{base, FN, peak}}}, \quad (11)$$

which nudges the inverter voltage until it matches the grid, so at closure there’s almost no jump. Turn it on with `presync_en`; the window runs from `t_presync` to grid connection `t_gridon`. It cut the inrush here by about  $25\times$  ( $\sim 7.6 \rightarrow 0.30$  p.u.).

## 6.2 Virtual resistance (damping)

The steps we used for simulation, ring the lightly-damped LCL filter. Instead of a real (lossy) resistor, we just subtract a fake resistive drop from the modulating voltage:

$$v_{\text{pwm}} = v_{\text{dVOC}} - R_v i_{\text{PCC}}. \quad (12)$$

$R_v$  (`Rv_damp`) damps the ringing like a lossless output resistor. The catch is a small voltage droop, so keep it modest (a few % of  $Z_{\text{base}}$ ); set `Rv_damp` = 0 to switch it off.

## 6.3 Integral $Q \rightarrow V$ loop (with anti-windup)

On its own, dVOC tracks  $Q$  with a *droop* offset, to nail  $Q$  exactly, an outer integrator trims the amplitude target:

$$V_{\text{ref}^2, \text{dyn}} = 2V_{\text{ref}}^2 + q_{\text{en}} \int K_{qv} (Q^* - Q) dt. \quad (13)$$

Turn it on with `q_en`.

# 7 The good and the not-so-good

What's great about it

- **No PLL** to go unstable on a weak or unbalanced grid, it makes its own angle.
- **Big-signal stable:** rides through large transients and *black starts* where small-signal droop would struggle.
- **Truly grid-forming:** it's a voltage source, so it can energise a network and support other units.
- **Acts like droop in steady state**, so it plays nicely with droop/GFM units on the same bus.
- **Two intuitive knobs:**  $\xi$  for amplitude speed,  $k_i$  for power speed.

What to watch out for

- **No built-in current limit.** As a voltage source it won't cap its own current; faults/large steps need a separate limiter (not in this model yet).
- **$Q$  needs help:** exact reactive tracking needs the extra integral loop, which needs anti-windup to stay safe.
- **Depends on  $\phi$ :** if  $\phi$  doesn't match the line angle,  $P$  and  $Q$  start to interfere each other.
- **LCL ringing:** the filter is lightly damped; virtual resistance helps but doesn't fully cure it.

# 8 Running it: the built-in test sequence

The model plays one 8 s story where each event is spaced out so you can watch it on its own. Every time and size below is just a variable in the init script, so you can move anything you like.

| Time (s)  | Event                        | What it shows                              |
|-----------|------------------------------|--------------------------------------------|
| 0         | Black start                  | Oscillator starts itself from a tiny seed. |
| 0.6 → 1.2 | Load on / off (+20/ − 20 MW) | Load steps while islanded.                 |
| 1.4       | Pre-sync window opens        | Inverter lines up with the grid.           |
| 2.0       | Grid connection              | Soft, near-zero-inrush close.              |
| 2.8       | $P^* \rightarrow 30$ MW      | Active-power tracking.                     |
| 3.8       | $Q^* \rightarrow 10$ MVar    | Reactive-power tracking.                   |
| 4.6 → 5.0 | Voltage sag (0.8 p.u.)       | Riding through a dip.                      |
| 5.6       | Frequency step (−0.2 Hz)     | Frequency-disturbance response.            |
| 6.2 → 6.6 | Phase jump (10°)             | Angle-disturbance response.                |
| 6.8       | $Q^* \rightarrow 0$          | Back to nominal before islanding.          |
| 7.2       | Islanding (breaker opens)    | Going autonomous.                          |
| 8.0       | End                          | —                                          |

## 8.1 dVOC core and interface

| Quantity                     | Variable          | Value / expression                    |
|------------------------------|-------------------|---------------------------------------|
| Amplitude speed (Hopf)       | <b>xi</b>         | 100                                   |
| RMS voltage reference (p.u.) | <b>Vref</b>       | 1                                     |
| Internal voltage gain        | <b>kv</b>         | 1                                     |
| Virtual capacitor / inductor | <b>Cosc, Losc</b> | 1, $1/\omega_g^2$                     |
| Characteristic impedance     | <b>eps</b>        | $\sqrt{L_{osc}/C_{osc}} = 1/\omega_g$ |
| Initial seed                 | <b>vc0</b>        | $10^{-3}$                             |
| Rotation angle               | <b>phi</b>        | $\pi/2$                               |
| Sync gain                    | <b>ki</b>         | 30                                    |
| Output voltage scaling       | <b>Kv_out</b>     | $V_{base, FN, peak}/\sqrt{2}$         |
| Current normalisation        | <b>Inorm</b>      | $I_{base, peak}$                      |

## References

- Rosso, R., Wang, X., Liserre, M., Lu, X., & Engelken, S. (2021). Grid-Forming Converters: Control Approaches, Grid-Synchronization, and Future Trends—A Review. *IEEE Open Journal of Industry Applications*, 2, 93–109. <https://doi.org/10.1109/OJIA.2021.3074028>
- Seo, G.-S., Colombino, M., Subotic, I., Johnson, B., Groß, D., & Dörfler, F. (2019). *Dispatchable virtual oscillator control for decentralized inverter-dominated power systems: Analysis and experiments* (Preprint NREL/CP-5D00-72274). National Renewable Energy Laboratory.